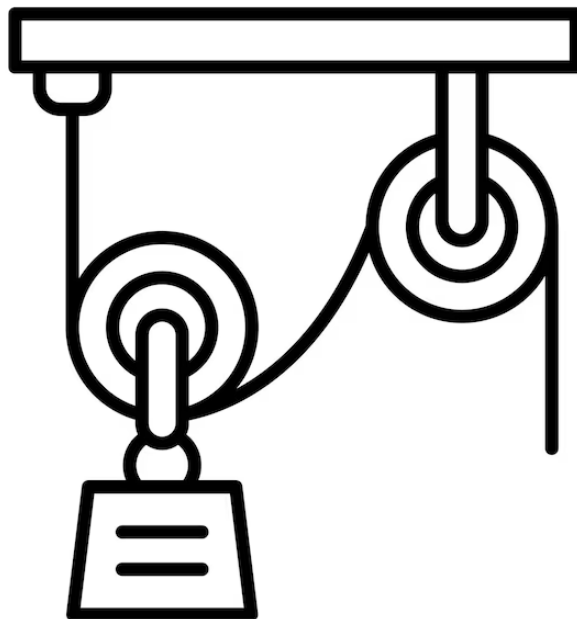


Science Olympiad
Science Olympiad at Berkeley Invitational

February 28, 2026

Machines B



Directions:

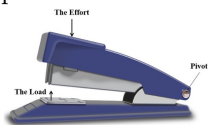
- Each team will be given **50 minutes** to complete the test.
- There are two sections: **§A** (MCQ), **§B** (Short Answer).
- For calculation questions, **use 3 or more sig. figs. in your answers** unless otherwise specified.
- Work will be graded, so it is recommended you show all work.
- You will be called up during the test portion to test your device.
- Assume $g = 9.8m/s^2$ and that friction and air resistance are negligible unless otherwise stated.
- Tiebreakers, in order: ES, TS, R1, C3, C2, C1, §B, §A.
- Good luck!

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Section A: Multiple Choice

This section consists of 28 multiple choice questions. The first 16 questions are worth 1 point. The last 12 are worth 2 points, for a total of 40 points. Each question has 1 correct answer.

1. A stapler as depicted below is what type of simple “machine”?



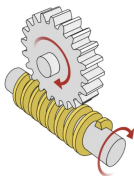
- A. Screw
B. Wedge
C. Inclined Plane
D. Second Class Lever
E. Third Class Lever
2. If the load on a lever is 12 N, and the ideal mechanical advantage of the lever at a particular point is 5, what force must be put on that point in theory to put the lever in equilibrium?

- A. 0.42 N D. 12 N
B. 2.4 N
C. 7 N E. 60 N

3. If the actual amount of force needed to put the lever in equilibrium is 10 N at that point, what is the actual mechanical advantage?

- A. 0.83 D. 2
B. 1.2
C. 1.5 E. 5

4. What type of gear is pictured below?



- A. Helical Gear
B. Spur Gear

- C. Rack and Pinion
D. Sun and Planet Gear
E. Worm Gear

5. What is one of the typical ADVANTAGES of the gear above?

- A. They are the ONLY gear type that allows for high gear ratios
B. They have a very high efficiency and very little friction due
C. They tend to be self-locking
D. They are slower and less efficient than rack and pinion gears at converting rotational to translational motion
E. None of these are advantages

6. What is one of the typical DISADVANTAGES of the gear above?

- A. They do not allow for high gear ratios
B. They have a very low efficiency due to very high friction
C. They are not self-locking
D. They are faster and more efficient than rack and pinion gears at converting rotational to translational motion
E. None of these are disadvantages

7. A basketball is thrown directly upwards with an initial velocity of 6 m/s. What is the maximum height it reaches? Assume no air resistance or friction.

- A. 1.24 m D. 3 m
B. 1.84 m
C. 2.24 m E. 6 m

8. How long does it take the basketball to reach this height?
- A. 0.61 s D. 1.84 s
 B. 0.8 s
 C. 1.24 s E. 2.24 s
9. What is the SI unit for work?
- A. m D. J
 B. s
 C. E E. W
10. What is the kinetic energy of a 2kg mass moving at 10 cm/s?
- A. 0.01 J D. 100 J
 B. 1 J
 C. 20 J E. 200 J
11. Two gears are meshed together and are currently spinning. Over time, they come to a stop. What is happening here?
- A. Their kinetic energy is being converted into other forms of energy like heat and sound
 B. Their kinetic energy is being converted into gravitational potential energy
 C. Their potential energy is being converted into other forms of energy like heat and sound
 D. Their potential energy is being converted into kinetic energy
 E. The law of energy conservation is being broken
12. If the gears continue spinning forever at the same speed, what does this tell you about the system?
- A. The gears are spinning infinitely fast
- B. There is no conversion of kinetic energy to any other form of energy
 C. Only kinetic energy is being converted, but not potential energy
 D. Their ideal mechanical advantage is 0
 E. Their ideal mechanical advantage is infinite
- Refer to this scenario for the next 4 questions: Connor is opening a trapdoor in the floor of uniform density. He is pulling straight upwards from the handle at the edge of the trapdoor. The trapdoor weighs 2 kg, is 0.8 meters long, and its hinge has zero friction.
13. What class of lever is this?
- A. First Class Lever
 B. Second Class Lever
 C. Third Class Lever
 D. Fourth Class Lever
 E. It's not a lever
14. The smallest force Connor must exert on the trapdoor handle to start opening it is at least:
- A. 4.9 N
 B. 9.8 N
 C. 14.7 N
 D. 19.6 N
 E. Connor cannot open the trapdoor in this configuration
15. If the minimum force required for Connor to start opening the trapdoor is 12 N, what is his actual mechanical advantage?
- A. 0.61 D. 1.63
 B. 0.8
 C. 1.25 E. 2

16. Assume Connor continues to pull perpendicular to the trapdoor as shown in Case 2. Once the trapdoor has been opened 45 degrees, has Connor's actual mechanical advantage changed?
- Yes, it has INCREASED since the ratio between the torque he exerts and the torque created by the trapdoor's gravity has INCREASED
 - Yes, it has INCREASED since the ratio between the torque he exerts and the torque created by the trapdoor's gravity has DECREASED
 - Yes, it has it has DECREASED since the ratio between the torque he exerts and the torque created by the trapdoor's gravity has INCREASED
 - Yes, it has it has DECREASED since the ratio between the torque he exerts and the torque created by the trapdoor's gravity has DECREASED
 - No, it remains the same since mechanical advantage is the ratio between lever arms, not torque
17. If a car wanted to increase its kinetic energy by a factor of 3, what could it do?
- Decrease its mass by 1.7x
 - Decrease its speed by 3x
 - Increase its mass by 1.7x
 - Increase its speed by 1.7x
 - Increase its speed by 3x
18. What is the ideal mechanical advantage of a gear ratio between a gear of 60 teeth and one of 12 teeth?
- 0.5
 - 2.5
 - 5
 - 6
 - 12
19. An inclined plane with an angle of 45 degrees to the ground has a 30 kg block resting on it. There is a force pushing the block up and parallel to the slope of 100 N. What is the lowest the coefficient of friction could be to prevent the block from sliding down?
- 0.26
 - 0.42
 - 0.52
 - 0.71
 - 1
20. If the force pushing the block up the slope suddenly disappeared, how fast would the block accelerate downwards in m/s^2 ? Assume the coefficient of static and kinetic friction is identical.
- 1.0
 - 1.3
 - 2.0
 - 2.7
 - 3.3
- Refer to the following scenario for the next 3 questions: Aiden is skydiving and was just dropped from an altitude of 3000m. He weighs 100 kg.
21. How long will it take him to reach an altitude of 1200m? Assume no air resistance, no terminal velocity, and that he has not pulled his parachute yet.
- 12 s
 - 17 s
 - 19 s
 - 25 s
 - 41 s

22. If Aiden pulls his parachute at 1200m, and suddenly starts to be decelerated by 20 m/s^2 , what roughly is the force the parachute is providing upwards at that instant?

A. 490 N
 B. 1000 N
 C. 2000 N
 D. 3000 N
 E. 3980 N

23. If Aiden's speed decreases to 0 m/s the instant he touches the ground (altitude of 0m), what was the magnitude of his average acceleration between when he pulled the parachute (1200m) and when he touched the ground in m/s^2 ?

A. 4.9
 B. 6.0
 C. 9.8
 D. 12.2
 E. 14.7

24. Jake rolls a bowling ball with a mass of 4 kg and strikes a singular pin with an initial speed of 8 m/s . If the pin has a mass of 1.5 kg , and flies off with a speed of 10 m/s , what is the final speed of the bowling ball? Assume momentum is conserved.

A. 2.85 m/s
 B. 3.20 m/s
 C. 4.25 m/s
 D. 6 m/s
 E. 10 m/s

25. How much work did Jake put into the ball? Assume the ball did not lose any energy between when Jake released it and it struck the pin.

A. -128 J
 B. -64 J
 C. -16 J

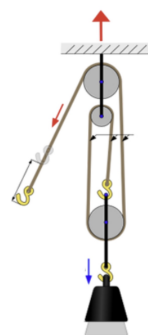
D. 111 J
 E. 128 J

26. Roughly how much energy was lost in the collision?

A. 11 J
 B. 13 J
 C. 17 J
 D. 36 J

E. The system actually gained energy

27. Danielle is pulling on a rope. The rope is connected to a pulley system as shown below with a 10 kg weight on the other side. How much force must be exerted on the rope to the left to keep the system in equilibrium? Assume an ideal system.



A. 2.5 N
 B. 3.3 N
 C. 28 N
 D. 32.6 N
 E. 98 N

28. If Danielle pulled on the rope with the force found above, but instead of being at equilibrium, the weight accelerated downwards at a rate of 1 m/s^2 , what is the actual mechanical advantage of the system?

A. 1.8
 B. 2.0
 C. 2.3
 D. 2.6
 E. 3.0

Section B: Short Answer

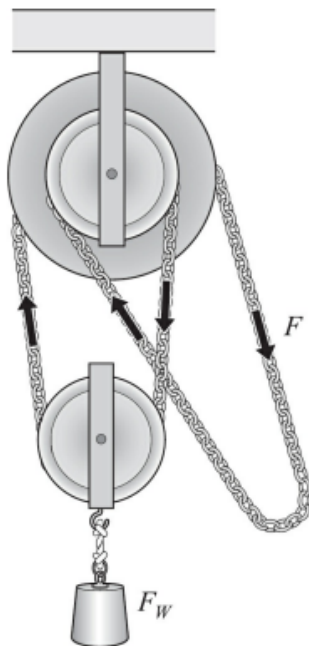
This section consists of 14 short answer questions. Points are shown for each question, for a total of 50 points.

1. [1 pt] The image in the test cover is a picture of what simple machine?
2. [2 pts] A wheel of radius 0.50 m is attached to an axle of radius 0.10 m. A force is applied to the wheel to lift a 400 N load attached to the axle. Friction causes a resisting torque equivalent to $40 \text{ N} \cdot \text{m}$. Find the ideal and actual effort forces.
3. [1 pt] A screw jack has a pitch of 4 mm. The handle has a radius of 0.25 m. Find the ideal mechanical advantage of the screw.
4. [1 pt] Why does increasing friction always reduce the efficiency of a simple machine?
5. [2 pts] What is the minor diameter of a screw that has a major diameter of 12 mm, a pitch of 1.5 mm, and a thread angle of 60° ?
6. [2 pts] Why does a third-class lever always have a mechanical advantage less than 1?
7. [3 pts] A crate of weight 600 N is pushed up an inclined plane that makes a 30° angle with the horizontal. The coefficient of kinetic friction between the crate and the plane is 0.20. What is the efficiency of the inclined plane?
8. [1 pt] The hull of a ship is actually a simple machine. What type of simple machine is it?
9. [2 pts] A pulley system has 6 supporting rope segments. A person lifts a 900 N load but must pull with 180 N. Find the efficiency of the system.
10. [2 pts] If a compound machine is created with 3 simple machines (with mechanical advantages X, Y, Z) in series, what is the mechanical advantage of the compound machine?
11. [2 pts] The load has a weight of 60 lbs. How many pounds should somebody exert on the system at the arrow to lift it? Refer to the image on the left.



12. [2 pts] The second and third image show what type of gear systems?
13. [2 pts] A uniform solid disk of mass M and radius R has a circular hole of radius $R/3$ drilled out. The hole's center is located at a distance of $R/2$ from the disk's center. Find the moment of inertia of the remaining shape about the original disk center.
14. [2 pts] A wedge with an angle 9° is used to lift a 6,000 N load. If the applied horizontal force is 1,500 N, what is the efficiency of this wedge?

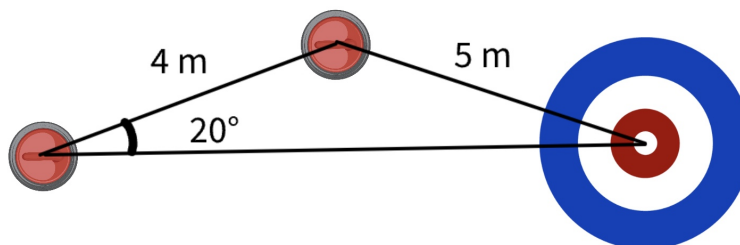
15. [12 pts] A differential pulley (Weston block) consists of two joined pulleys of different radii ($R = 15\text{ cm}$, $r = 12\text{ cm}$) on the upper block and one lower moveable pulley. Refer to the image below.



- (a) [2 pts] What is the IMA formula for a differential pulley and what is the IMA for this specific setup?
- (b) [2 pts] A 100 kg mass is suspended. If the system is in static equilibrium, what is the tension in the slack part of the chain in Newtons? (Assume 100% efficiency for this part).
- (c) [2 pts] 100% efficiency is pretty difficult to achieve. So let's say the system is actually 60% efficient. If an effort of 150 N is applied, calculate the maximum load the system can actually lift in Newtons.
- (d) [3 pts] While lifting a 50 kg load, the effort chain is pulled at a constant velocity of 0.5 m/s. What is the power output (useful power) of the system in watts?
- (e) [2 pts] If the load is raised 2 meters, calculate the change in gravitational potential energy of the load in joules.
- (f) [1 pt] Suppose the effort is suddenly released and the load falls. If the pulleys have significant inertia, will the acceleration of the load be equal to, less than, or greater than g ?

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16. [13 pts] With the Winter Olympics recently concluding in Italy (well, currently ongoing at the time of writing), let's take a look at an interesting sport, curling. Curling consists of "stones" (the circles in red) that you want to put in the center of the target to earn the most amount of points. Each stone has a diameter of 30 cm and a mass of 18 kg. It is Team USA's turn. All collisions are perfectly elastic. The stone on the left is the first stone.



- [2 pts] To begin the round, one of your muscles (your bicep) pushes the stone towards the target. If your muscle has an IMA of 0.15. The stone has a length of 10 cm and a resistive band length of 7 cm. What is the total IMA of the arm-handle/band system?
- [3 pts] Player A on team USA wants to hit stone A into B, which is currently located 20 degrees above the horizontal and the centers are 4 meters away. What is the minimum angle from horizontal at which stone A still hits stone B?
- [3 pts] Player B tells player A that the second stone needs to go completely horizontal. At what angle must the first stone be launched to cause the second stone to go completely horizontal?
- [2 pts] Team USA wants to aim in the exact center of the target. How many degrees under the horizontal should the second stone travel to be in the center? The second stone is 5 meters away from the dead center of the target.
- [3 pts] Instead of trying to hit the second stone, player A decides to go straight for the target. Ice has a very low coefficient of friction, such that we can say $\mu_k = \mu_s = 0.06$. For points to be scored, the stone must be completely inside a certain ring to earn the points for that ring. Otherwise, the lower value for points is taken. The inner circle has an outer diameter of 0.33 meters, the red circle has an outer diameter of 1 meter, the second white ring has an outer diameter of 2 meters, and the blue ring has a diameter of 3 meters. If Team USA wants to score in the outer white ring, what is the minimum force that should be applied to handle for the first stone to stop in the outer white ring? Assume a contact time of 1.0 seconds.